

Do Foreign Investors Perform Better Than Locals?

- Information Asymmetry, Investor Sophistication, and Market Liquidity -

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ABSTRACT

This study analyzes the extent to which information asymmetry, investor sophistication, and market liquidity affect the relative performance of different investor groups (foreigners, local institutions, and local individuals), using a daily transaction data from the Korea Stock Exchange over a four-year period. Empirical findings show that, in large stocks, where information asymmetry problem is small, foreigners outperform locals, reflecting the degree of investor sophistication. In small stocks, where information asymmetry problem is grave, however, foreigners turned out to under-perform locals. Findings also show that, due to market liquidity, foreigners are disadvantaged when trading large chunk of shares. The popular strategy of tracking foreigners also turned out to have greatest profitability in large stock, where information asymmetry problem is small.

Key words: information asymmetry, investor sophistication, and market liquidity

JEL classification: G150

This paper is motivated by seemingly conflicting results in the existing literature on the relative performance of foreign and local investors. One set of studies shows evidence that locals outperform foreigners. Shukla and Inwegen (1995) demonstrate that U.K. open-end mutual funds (foreigners) under-perform U.S. open-end mutual funds (locals) when both group of managers are investing in the U.S. market. Hau (2001) shows that European traders at non-German speaking locations under-perform Frankfurt-based traders when both group of investors are trading German stocks. Choe, Kho, and Stulz (2001) demonstrate that foreign investors are significant net buyers before a negative abnormal return event and significant net sellers before a positive abnormal return event.

Contrary to such findings, a set of studies tells us an opposite story. Grinblatt and Keloharju (2000), studying the Finish stock market, find that foreigners buy more stocks that perform well in the future than domestic investors. Seasholes (2000) show that foreigners systematically accumulate shares before positive earning announcements and sell-off shares before negative earning announcements. Froot, O'Connell, and Seasholes (2001), Bekaert, Harvey, and Lumsdaine (1999), and Seashles (1999) find positive correlation between foreign equity flows and future returns, indirectly indicating that foreign investors perform better than locals.

Such seemingly conflicting results in the existing literature, however, can be resolved once we acknowledge that more than one factor is at work in determining the relative performance of foreign and local investors. In this paper, I argue that three factors are at work: information asymmetry, investor sophistication, and market liquidity. As such, a paper capturing mostly the information asymmetry effect tends to show that locals outperform foreigners. On the other hand, if a paper captures mostly the effect of investor sophistication, the paper tends to show that foreigners outperform locals.

The works of Shukla and Inwegen (1995) and Hau (2001), for example, can be understood as studies that capture only the information asymmetry effect on performance. By comparing U.S. and U.K. mutual funds, which can be seen to have the same level of investor sophistication, Shukla and Inwegen (1995) in effect controls for the difference in investor sophistication. Also, by studying the U.S. market, which is the most liquid market in the world, they control for market liquidity. As a result, they exclusively capture the effect of information asymmetry on performance. Hau (2001), likewise, compares European traders in different locations, which can be seen to have the same

degree of investor sophistication. By doing this, he in effect controls for the difference in investors sophistication. Also, by studying the German market, which can be more or less regarded as liquid, controls for market liquidity. As a result, he also exclusively captures the effect of information asymmetry on performance.

On the other hand, the works of Grinblatt and Keloharju (2000) and Seasholes (2000) can be understood as studies that capture only the effect of investor sophistication on relative performance. By focusing on the largest 16 Finnish stocks, which are liquid and have sufficient amount of information even for foreign investors, Grinblatt and Keloharju (2000) in effect control for market liquidity and information asymmetry and exclusively capture the effect of investor sophistication on performance. Seasholes (2000), likewise, focuses on Taiwanese firms with market capitalization over USD 1 billion and 14 largest companies in Korea, where sufficient amount of information is available even for foreign investors. By doing this, he in effect controls for information asymmetry. Also, by looking at earning announcements instead of stock returns, he effectively controls for market liquidity. As a result, he also exclusively captures the effect of investor sophistication on performance.

Third, the works of Choe, Kho, and Stulz (2001), Froot, O'Connell, and Seasholes (2001), Bekaert, Harvey, and Lumsdaine (1999), and Seasholes (2000) can be understood as studies that capture a combined effect of all three factors. Relatively speaking, however, Choe, Kho, and Stulz (2001) can be understood as mostly capturing either the information asymmetry effect or the market liquidity effect. By treating each individual stock equally, regardless of their size, Choe, Kho, and Stulz (2001) analyses a sample where small stocks outnumber large ones. Small stocks can be presumed to be more subject to information asymmetry and market liquidity problems. The other three works, on the other hand, use value-weighted market indices in their analyses and therefore can be understood as mostly capturing the effect of investor sophistication.

Such reasoning, however, is ad hoc. The studies in the existing literature adopt different performance measures, different data frequencies, different investment horizons, and look at different markets, which inevitably blur the comparison of the results and make it hard to identify general patterns.¹ As such, in this paper, I attempt to analyze the

¹ For example, as for a performance measure, Shukla and Inwegen (1995) look at Sharpe's ratio, Treynor's ratio, Jensen's alphas, and timing ability, while Grinblatt and Keloharju (2000) compares

influence of each three factor under a unified framework. That is, using a common performance measure, common data frequencies, and common investment horizons, in a common market. The empirical works in this paper are organized around the following four hypotheses.

- H1. *Investor sophistication*: In large stocks, after controlling for liquidity, foreign investors outperform locals.
- H2. *Information asymmetry*: In small stocks, after controlling for liquidity, foreign investors outperform locals less significantly or under-perform locals.
- H3. *Market liquidity*: Foreign investors that trade more than 1% of outstanding shares in a day under-perform those foreign investors trading less, when information asymmetry is controlled for.
- H4. *Tracking foreign investors*: The profitability of tracking foreigners is greatest in large stocks, where information asymmetry is small.

The above hypotheses implicitly assume that size can effectively control for information asymmetry. I believe this is a reasonable assumption. Large stocks tend to have more analyst coverage, including foreign analysts, and tend to be listed as ADRs or GDRs in foreign stock exchanges, making them familiar even to foreign retail investors.

Related to the belief that foreigner investors are more sophisticated than locals, in Hypothesis 4, I also test the profitability of tracking foreigners.² That is, I explore whether one can obtain abnormal returns by buying those stocks foreigners have recently purchased and selling those that foreigners have recently dumped. If foreign investors outperform locals in large stocks, I expect to find the profitability of tracking foreigners to be greatest in large stocks as well.

The organization of this paper is as follows. Section I describes the data used in this paper. Section II explains the methodologies and Section III reports the results. Section

the difference of buy ratios in winning and losing stocks. As for data frequency, Shukla and Inwergen (1995) use monthly frequency, while Grinblatt and Keloharju (2000) use daily frequency. As for investment horizon, Shukla and Inwergen (1995) use a one-month horizon, while Grinblatt and Keloharju (2000) use a six-months horizon.

² This strategy has been very popular among the Korean locals, but to my knowledge, has not yet been carefully studied with actual data. A remotely related paper is Myers, Poterba, Shackelford, and Shoven (2001), which study the profitability of copycat funds.

IV concludes.

I. Data

This paper makes use of three data sets: daily transaction data and book value of equity data obtained from KSE (Korea Stock Exchange) and daily return data acquired from KSRI (Korea Securities Research Institute). The former data set covers the universe of trades that took place at KSE (Korea Stock Exchange) between December 1996 and December 2000.³ I did not collect information on options trading on individual stocks, since it was simply not allowed during the sample period.⁴ Also, I did not collect information on depository receipts since the main interest of this paper is to see how foreign investors perform when trading in a remote marketplace.

Figure 1 shows that, during this four-year period, KSE underwent many phases of the market. It first experienced a relatively tranquil period that continued until June 1997, then a market crash from late 1997 to early 1998, a market rally between June 1998 and December 1999, and lastly another market crash throughout year 2000. As such, a performance measure calculated over this four-year period, should be understood as a robust measure to market movements, such as rallies or crashes.

Specifically, the transaction data set contains following information for each common equity at a daily frequency: (i) investor type (domestic individual, domestic institution, or foreign) engaged in transaction, (ii) number of shares purchased, (iii) number of shares sold, (iv) number of outstanding shares, and (v) price at which the transaction was made. Since I am using a transaction data, there is no information on holding period for each stock. The book value of equity is obtained to compute book-to-market ratios. The book value of equity used in this paper, however, includes preferred stocks. Given the minimal proportion of preferred stocks in the Korean stock market, however, I do not

³ Other papers in the literature vary in their length of sample period. Hau (2001) covers only 4 months. Grinblatt and Keloharju (2001) and Choe, Kho, and Stulz (2001) cover 2 years of data. Froot, O'Connell, and Seasholes (2001) cover 4 and Seasholes (2000) covers 5 years. Sukula and Inwegen (1995) make use of 12 years of data, which is an exception in this literature.

⁴ Short selling was allowed, but heavily restricted.

believe this makes much of a difference.⁵ Also, I dropped observations when the book values are negative.⁶

The KSRI data set, on the other hand, provides daily return data adjusted for dividends, stock splits, and others. It is worth noting that all three data sets are free from survivorship bias. That is, they contain information even on those listings that existed in the past but no longer available at the time of data construction. To make a cross-group comparison valid, listings are dropped from the sample whenever foreign ownership ceiling was exhausted.⁷ Notice that, at the ceiling, foreigners can only sell even when a purchase is expected to be profitable. Information on the daily exhaustion of foreign ownership ceiling is obtained from KOSCOM (Korea Securities Computer Corporation).

In Table I, two measures of trading volume are computed for each investor group: (i) trading volume per stock-day and (ii) number of stocks traded per day. To compute trading volume per stock-day, absolute value of net investment is obtained for each stock-day, averaged across all stocks, and then averaged over all sample days. To compute number of stocks traded per day, number of traded stocks are counted at a given trading day and then averaged over all sample days. The table also reports each measure of trading by size quintiles. Numbers in Table 1 suggest that domestic individuals in aggregate trade in small amounts in a large number of listings, while foreign investors in aggregate trade in large amounts in a small number of listings. The table also shows that domestic institutions and foreign investors trade heavily in large stocks, while domestic individuals trade evenly across all size quintiles. Figure 1 depicts KOSPI and net investment by investor groups over the sample period.

⁵ For example, in the manufacturing sector, the total book value of preferred stocks took up only 0.7% of total book value of equity in fiscal year 2000.

⁶ Firms with negative book value of equity took up only 4% of the original sample.

⁷ For example, between May and November 1997, foreign investors, in aggregate, could not own more than 23% of the outstanding shares per company and foreign investors, individually, could not own more than 6%. Since May 1998, in principle, there exists no restriction on foreign ownership. Upper ceiling on foreign investors in aggregate changed from 10% (Jan, 1992) → 12% (Dec, 1994) → 15% (Jul, 1995) → 18% (Apr, 1996) → 20% (Oct, 1996) → 23% (May, 1997) → 26% (Nov, 1997) → 55% (Dec, 1997) → 100% (May, 1998). As for individual foreign investor, the upper ceiling changed from 3% (Jan, 1992) → 4% (Apr, 1996) → 5% (Oct, 1996) → 6% (May, 1997) → 7% (Nov, 1997) → 50% (Dec, 1997) → 100% (May, 1998). Source: Financial Supervisory Service (2000), page 1.

II. Methodologies

A. Measure of Performance

It is worth noting the nature of investment performance this paper is interested in. According to any finance textbook, overall performance can be attributed to two major investment decisions: asset allocation and securities selection. Given the restricted nature of the data set, this paper focuses only on securities selection (stock-picking, to be exact).⁸ It should also be noted that the performance measure used in this paper captures only buying and selling decisions, but leaves out the holding decision.⁹ To capture the effect of holding decisions, one needs to have a holding data.

For each of the three investor-groups, three performance measures are computed: (i) stock-picking, (ii) stock-buying, and (iii) stock-selling skills.¹⁰ To compute the stock-buying skill measure, at a given trading day, ex-post buy-and-hold returns are obtained for each stock that experienced a net investment. Then, value-weighted average of these returns are computed for each trading day. Weights are computed to reflect the relative size of each net investment. Ex-post returns are obtained for four investment horizons: a month, a quarter, a half-year, and a year.¹¹ Notice that investment horizons do not go beyond a year. This makes the performance measure less prone to the biases mentioned by Barber and Lyon (1997).¹²

⁸ To measure performance due to asset allocation, one should have information on the equity holdings in other exchanges and holdings in other asset categories. Moreover, given that index futures and options are heavily used as cost-saving methods to change asset allocation, trading information on such derivative instruments are also needed.

⁹ See Chen, Jegadeesh, and Wermers (2000) for measuring performance without a holding data.

¹⁰ Notice that option trading on individual stocks was not allowed by the exchange during the sample period.

¹¹ Returns over each investment horizon are respectively obtained by compounding daily returns over 24, 72, 144, and 288 trading days. Note that KSE is open 6 days a week.

¹² Barber and Lyon (1997) document that long-run abnormal returns, calculated using reference portfolios, such as an equally weighted market index, yield biased test statistics for three reasons: new listing, rebalancing, and skewness biases.

$$SB(i, n, t) = \sum_{j=1}^J w(i, j, t) R(j, n, t) \text{ where } w(i, j, t) = \frac{NI(i, j, t)}{\sum_{j=1}^J NI(i, j, t)} \quad (1)$$

$SB(i, n, t)$ measures stock-buying skill of investor group i at trading day t over a n -day investment horizon. $R(j, n, t)$ denotes ex-post buy-and-hold return of stock j at day t for an n -day investment horizon.¹³ $NI(i, j, t)$ denotes net investment into stock j at day t .

To compute the stock-selling skill measure, at a given trading day, ex-post buy-and-hold returns are obtained for each stock that experienced a net divestment. Then, value-weighted average of these returns are computed for each trading day. This is then multiplied by -1 so that superior stock-selling skill takes a positive value. Weights are computed to reflect the relative size of each net divestment.

$$SS(i, n, t) = -\sum_{j=1}^J w(i, j, t) R(j, n, t) \text{ where } w(i, j, t) = \frac{ND(i, j, t)}{\sum_{j=1}^J ND(i, j, t)} \quad (2)$$

$SS(i, n, t)$ measures stock-selling skill of investor group i at trading day t over a n -day investment horizon. $ND(i, j, t)$ denotes net divestment from stock j at day t .

To compute the overall stock-picking skill measure, at a given trading day, value-weighted average of ex-post buy-and-hold returns are computed for two groups of stocks: stocks that were net-invested and stocks that were net-divested. The two value-weighted returns are then combined by a weight that reflects the relative size of total net investment and total net divestment.

$$SP(i, n, t) = wSB(i, n, t) + (1 - w)SS(i, n, t) \quad (3)$$

$$w(i, j, t) = \frac{\sum_{j=1}^J NI(i, j, t)}{\sum_{j=1}^J NI(i, j, t) + \sum_{j=1}^J ND(i, j, t)}$$

¹³ Notice that I am using KRW returns even for foreign investors. This is because I am interested only in stock-picking skills. Speculation in the FX market should be understood as a separate skill.

$$SP(i, n) = \frac{1}{T} \sum_{t=1}^T SP(i, n, t) \quad (4)$$

$SP(i, n, t)$ measures stock-picking skill of investor group i at trading day t , over an n -day investment horizon, while $SP(i, n)$ measures stock-selling skill of investor group i for an n -day investment horizon.

B. Investment Style Adjustment

Performance measures explained above, however, do not control for stock characteristics. Depending upon the investment style a trader adopts, high performance can be achieved even without any stock-picking ability. Thus, returns in equations (1) and (2) are adjusted by a matching portfolio return obtained à la Daniel, Grinblatt, Titman, and Wermers (1997).¹⁴ First, all listings are ranked by their market capitalization, from which quintile portfolios are formed. Second, each quintile is further divided into sub-quintiles based on book-to-market ratio.¹⁵ This two-dimensional grouping procedure results in 25 portfolios with distinct combination of size and book-to-market ratio characteristics. Return from each matching portfolio is computed by taking a value-weighted average of ex-post buy-and-hold returns on those stocks comprising the portfolio.¹⁶ The two-dimensional grouping procedure is repeated for each trading day.

$$SB^*(i, n, t) = \sum_{j=1}^J w(i, j, t) [R(j, n, t) - R^*(j, n, t)] \quad (1')$$

$$SS^*(i, n, t) = -\sum_{j=1}^J w(i, j, t) [R(j, n, t) - R^*(j, n, t)] \quad (2')$$

¹⁴ Barber and Lyon (1997) show that such matching portfolio approach yields well-specified test statistics even when testing long-run abnormal returns. Lyon, Barber, and Tsai (1999) document that such matching portfolio approach can be further improved by using a bootstrapped skewness-adjusted test statistic or empirical p-values. Such improved methods are not adopted in this paper since I do not deal with investment horizons beyond a year.

¹⁵ Momentum effect is not controlled for since, in this paper, I regard momentum or value strategy as an important factor determining investment performance.

¹⁶ Here, notice that I am using buy-and-hold returns (BHR) instead of cumulative abnormal returns (CAR). CAR is not used since it computes only the mean daily abnormal returns, regardless of the investment horizon.

$SB^*(i, n, t)$ and $SS^*(i, n, t)$ respectively denote stock-buying and stock-selling skills of investor group i at trading day t over an n -day investment horizon. $R^*(j, n, t)$ denotes the n -day return of a matching portfolio of which stock j is a member at day t .

C. Tracking Strategy

To test the profitability of a tracking strategy, three measures are used to identify those stocks popular (or unpopular) among foreign investors: (i) net purchase, (ii) net investment, and (iii) net purchase ratio. Net purchase is defined as number of shares purchased minus number of shares sold. Net investment is defined as net purchase multiplied by share price. Net purchase ratio is defined as net purchase divided by number of shares outstanding.

$$NP(j, t) = [\text{no. of shares purchased}] - [\text{no. of shares sold}] \text{ on stock } j \text{ at day } t \quad (5)$$

$$NI(j, t) = NP(j, t) \times P(j, t) \quad (6)$$

$$NPR(j, t) = NP(j, t) / [\text{no. of shares outstanding of stock } j \text{ at day } t] \quad (7)$$

KSE uses the first two methods. That is, it reports top-20 and bottom-20 stocks according to net purchase and net investment every day through its website.¹⁷ However, these are biased measures because large stocks are more likely to be included in the top-20 and the bottom-20 stocks. This prompted me to add the third measure, which is not biased toward large stocks.

Using each method, two portfolios are formed for each trading day: a portfolio comprising top-20 stocks (top-portfolio) and a portfolio comprising bottom-20 stocks (bottom-portfolio). Value-weighted ex-post buy-and-hold return is then computed for each portfolio.¹⁸ Strategies *I*, *II*, and *III* go long with the top-portfolio and go short with the bottom-portfolio. Each strategy respectively uses net purchase, net investment, and net purchase ratio to identify the top-and the bottom-portfolios. Return from each of these strategies is computed by combining the returns on the two portfolios with a weight that reflects the relative size of the two. Strategies *IV*, *V*, and *VI*, on the other hand,

¹⁷ KSE also reports top-20 and bottom-20 gross buys and gross sells by foreigners, which I do not analyze here. See www.kse.or.kr

¹⁸ This implies that, to track foreigners, an investor must invest in top-20 stocks according to the relative size of each component stocks and divest from bottom-20 stocks according to the relative size of each component stocks.

merely go long with the top-portfolios. Each strategy respectively uses net purchase, net investment, and net purchase ratio to identify the top-portfolios.

III. Results

A. Performance Comparison

Table 2 compares investment performance (stock-picking skills) among three investor-groups. Notice that performance is measured over a one-month horizon and adjusted for stock characteristics. The table is divided into two dimensions: size quintiles (in rows) and market liquidity (in columns). Notice that size quintiles are updated each trading day and market liquidity is represented by number of shares purchased or sold by foreigners over total number of outstanding shares. Market is judged to be illiquid if the ratio is greater than 0.01 (1%). Standard errors are in the parentheses and numbers of observations (number of trading days) are in the brackets.

Three points stand out from the table. First, foreign investors outperform locals in the largest size quintiles when foreigners do not trade in large chunks. Foreign investors outperform the benchmark portfolio by 20 basis points, while local individuals and institutions under-perform by 80 basis points (see columns 4, 5, and 6 in the largest size quintiles). The differences are also statistically significant at 1% level (see columns 7 and 8 in the largest size quintiles). Such findings confirm hypothesis H1 introduced in the earlier part of this paper. Since information asymmetry and market liquidity are controlled for, the finding suggests that foreign investors are more sophisticated than locals.

Second, local investors outperform foreigners in the smallest size quintiles when foreigners do not trade in large magnitudes. Local investors outperform the benchmark portfolio by 60-70 basis points, while foreign investors under-form by 110 basis points (see columns 4, 5, and 6 in the smallest size quintiles). The differences are also statistically significant at 1% level (see columns 7 and 8 in the smallest size quintiles). Such findings confirm hypothesis H2 introduced earlier. It implies that the effect of information asymmetry dominates that of investor sophistication in small stocks.¹⁹

¹⁹ The finding that foreign investors' relative profitability is positively correlated with firm size is consisted with the finding in Seasholes (2000).

Third, foreign investors under-perform when trading in large magnitudes. When trading more than 1% of outstanding shares in a day, they under-perform the benchmark portfolio by 210 basis points (see column 3 in the largest size quintiles). When trading less than 1%, however, they outperform by 20 basis points (see column 6 in the largest size quintiles). The difference is also statistically significant at 1% level (see column 9 in the largest size quintiles). Such finding supports hypothesis H3 stated earlier.

B. Robustness Checks

The results are robust to investment horizon. Tables 3, 4, and 5 show results using longer investment horizons: quarter, half-year, and year. All three points stylized in Table 2 applies to other investment horizons. That is, in large stocks, after controlling for liquidity, foreign investors outperform locals. In small stocks, after controlling for liquidity, local investors outperform foreigners. Foreign investors that trade more than 1% of outstanding shares in a day under-perform those foreign investors trading less, when information asymmetry is controlled for.

Thus far, I compared mean performances among investor-groups. Given the likely skewness of returns, in Table 6, I also compare median performances. The basic patterns do not change.

It is also worth noting here why I am treating each trading day's return as an independent observation. To purchase shares, a typical investor may purchase them in piecemeal so as not to be disadvantaged by market illiquidity. So, when working with ex-post buy-and-hold returns, say over a month horizon at a daily frequency, returns will be overlapping and one should definitely worry about the independence of each observation.

However, in my paper, I am dealing with an 'investor group,' not an 'individual investor.' As long as the trading pattern of each investor in given group is not perfectly correlated, the problem of overlapping returns is less worrisome. That is, the portfolio that investor group i is purchasing on day $t+1$ is unlikely to be the same portfolio that investor group i is purchasing on day t . That stocks being purchased and their weights in the portfolios are likely to be very different.

C. Profitability of Tracking Strategy

Table 6 reports the profitability of tracking foreign investors. That is, can one make an

abnormal return by buying stocks foreigners have recently bought and selling those foreigners have recently dumped? Results are shown for six tracking strategies. Strategies *I*, *II*, and *III* take a long position in top-20 stocks only. They do not take short positions.²⁰ Each strategy respectively uses net purchase (Strategy *I*), net investment (Strategy *II*), and net purchase ratio (Strategy *III*) to identify the top 20 stocks. On the other hand, Strategies *IV*, *V*, and *VI* take a long position in top-20 stocks, while taking a short position in bottom-20 stocks.

Results indicate that Strategy *VI*, long-cum-short strategy using net purchase ratio as an identifier, gives positive excess returns regardless of investment horizon. Returns from tracking are 53, 160, 232, and 323 basis points respectively for monthly, quarterly, half-yearly, and yearly horizons.²¹ To see if the tracking strategy is most profitable in large stocks, Strategy *VI*, is further analyzed across different size quintiles. Table 7 reports the results. Consistent with the earlier finding of foreign investors superior performance in large stocks (hypothesis H1), tracking strategy turns out to be most profitable in the largest size quintile.

IV. Conclusion

Empirical findings of this paper show that, in large stocks, where information asymmetry problem is relatively small, foreign investors turned out to outperform local individuals, as expected, reflecting the fact that foreign investors are more sophisticated. But, a more interesting finding is that, in small stocks, where information asymmetry problem is grave, foreign investors under-performed local investors, reflecting the fact that information asymmetry effect outweighs the effect of investor sophistication. Findings also show that, due to market liquidity, foreigners are disadvantaged when trading large block of shares. The results are robust to the types of local investors (individual versus institutions) and investment horizons (month, quarter, half-year, and year).

²⁰ Long strategies alone are separately investigated because of the legal constraints individual investors face on short selling.

²¹ One may argue that this finding is not consistent with the result that foreign investors under-perform the benchmark when it trades in large magnitudes. This is, however, not necessarily the case. This is because the threshold used in the earlier analysis to identify large trades (1% of total outstanding shares) is more restrictive than the identifier used to capture foreign popularity (top 20 and bottom 20 stocks).

The popular investment strategy of tracking foreigners turned out to be a profitable one. Even when stock characteristics, such as size and momentum, are controlled, buying stocks foreign investors recently purchased and selling those they recently dumped paid off. Also, such tracking strategy turned out to be most profitable in large stocks, where information asymmetry is relatively small. This finding is consistent with the earlier finding that foreign investors generally outperform locals in large stocks.

The paper, however, is not without limitations. First, given the restricted nature of the data set, as a measure of investment performance, this paper focuses only on securities selection (stock-picking, to be exact) and leaves out asset allocation decisions. Second, it should also be noted that the performance measure used in this paper captures only buying and selling decisions, but leaves out the holding decision. To capture the effect of holding decisions, one needs to have a holding data. Third, this paper does not make use of information on short selling of individual stocks, which was restrictively permitted during the sample period. Fourth, this paper uses transaction data, instead of order-flow data, in identifying liquidity constraints. Notwithstanding all such limitations, however, strong and consistent results found throughout this paper give a high degree of confidence over the assertions made in this paper.

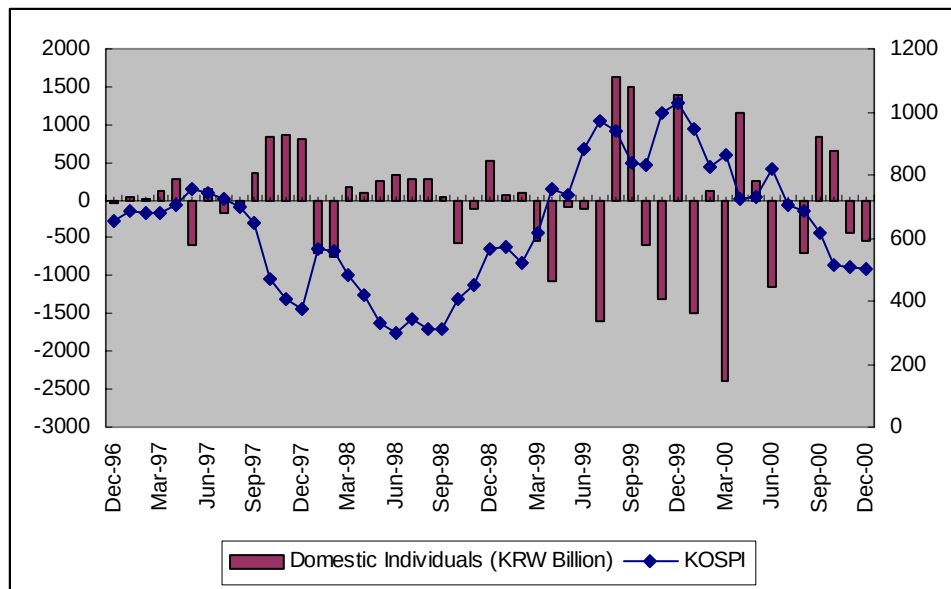
As an extension of this paper, it would also be meaningful to study who tracks whom. This paper identified how profitable it is to track foreigners. But, it did not show that local investors actually do track foreigners. One way of analyzing this is to use a Vector Autoregressive (VAR) model following Lee, Lin, and Liu (2000).

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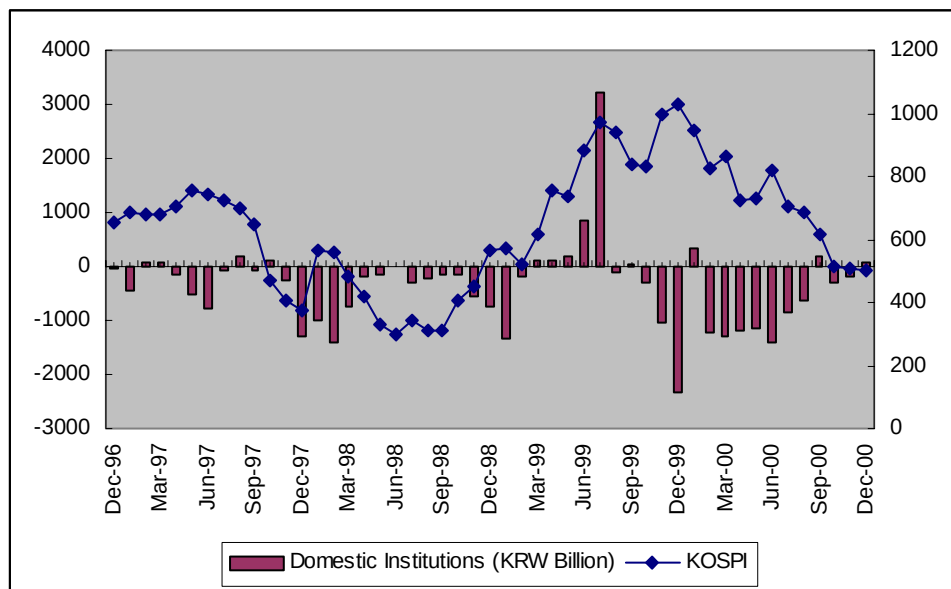
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Figure I
KOSPI and Net Investment by Investor Groups

a. Domestic Individuals



b. Domestic Institutions



c. Foreign Investors

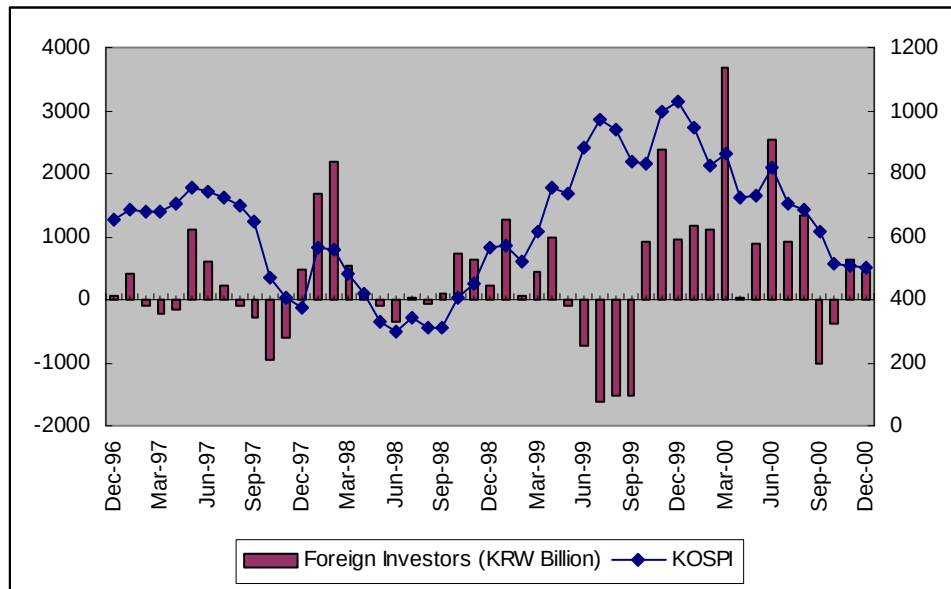


Table 1
Summary Statistics on Trading

Two measures of trading volume are computed for three investor groups. The two measures include (i) trading volume per stock-day and (ii) number of stocks traded per day. The three investor-groups include domestic individual investors, domestic institutional investors, and foreign investors. To compute trading volume per stock-day, absolute value of net investment is obtained for each stock-day, averaged across all stocks, and then averaged over all sample days. To compute number of stocks traded per day, traded stocks are counted per day and then averaged over all sample days. The table also reports each measure of trading by size quintiles. The sample includes all common stocks listed in the Korea Stock Exchange (KSE) from December 1996 to December 2000. Observations are dropped when book value of equity is negative or foreign ownership ceiling is exhausted.

	Domestic Individual	Domestic Institutions	Foreign Investors
Trading Volume per Stock-Day (Million Won)	295	369	690
Size 1 (small)	9	24	25
Size 2	32	53	46
Size 3	69	91	70
Size 4	147	162	109
Size 5 (large)	1230	1070	1270
Number of Stocks Traded per Day (#)	600	407	167
Size 1 (small)	120	41	6
Size 2	121	66	12
Size 3	121	85	22
Size 4	120	101	41
Size 5 (large)	119	113	85

Table 2
Performance Comparison
(Investment Horizon = one-month)

Table compares stock-picking skills among three investor-groups. Stock-picking skill is measured as a value-weighted average of stock-buying and stock-selling skills (weight = net investment and net divestment). Stock-buying skill, in return, is measured as a value-weighted average of ex-post buy-and-hold returns on stocks purchased by an investor group (weight = KRW amount of each stock's net investment). Stock-selling skill is measured as a value-weighted average of ex-post buy-and-hold returns on stocks sold by an investor group (weight = KRW amount of each stock's net divestment). Notice that performance is measured over a one-month horizon and adjusted for stock characteristics (size and book-to-market ratio). The table is divided into two dimensions: size quintiles (in rows) and market liquidity (in columns). Size quintiles are updated each trading day and market liquidity is represented by number of shares purchased or sold by foreigners over total outstanding number of shares. Market is judged to be illiquid if the ratio is greater than 0.01 (1%). Standard errors are in the parentheses and numbers of observations are in the brackets.

Size Quintile	(no. of shares purchased or sold by foreigners) / (outstanding no. of shares)						Difference in Mean Tests		
	greater than 1%			lower than 1%			[7] = [6] – [4]	[8] = [6] – [5]	[9] = [3] - [6]
	[1] individual	[2] institution	[3] foreigner	[4] individual	[5] institution	[6] foreigner			
1 (small)	-0.017 (0.019) [162]	-0.002 (0.026) [96]	-0.020 (0.019) [164]	0.006** (0.004) [1072]	0.007** (0.004) [1072]	-0.011** (0.006) [1041]	-0.018*** (0.007)	-0.018*** (0.008)	
2	0.007 (0.014) [307]	0.015 (0.017) [247]	0.007 (0.013) [307]	-0.002 (0.002) [1072]	-0.003 (0.002) [1072]	-0.001 (0.004) [1069]	0.002 (0.004)	0.002 (0.004)	
3	0.006 (0.013) [352]	0.002 (0.010) [331]	0.007 (0.013) [353]	-0.002 (0.002) [1073]	-0.002 (0.002) [1073]	-0.003 (0.003) [1071]	-0.001 (0.004)	-0.001 (0.004)	
4	0.000 (0.011) [417]	-0.001 (0.011) [408]	-0.002 (0.010) [417]	-0.011*** (0.001) [1072]	-0.011*** (0.001) [1072]	0.005*** (0.002) [1072]	0.017*** (0.002)	0.017*** (0.002)	
5 (large)	-0.017** (0.009) [529]	-0.021*** (0.009) [529]	-0.021*** (0.009) [530]	-0.008*** (0.001) [1072]	-0.008*** (0.001) [1072]	0.002** (0.001) [1072]	0.010*** (0.002)	0.010*** (0.002)	-0.023*** (0.009)

Table 3
Performance Comparison
(Investment Horizon = one-quarter)

Table compares stock-picking skills among three investor-groups. Stock-picking skill is measured as a value-weighted average of stock-buying and stock-selling skills (weight = net investment and net divestment). Stock-buying skill, in return, is measured as a value-weighted average of ex-post buy-and-hold returns on stocks purchased by an investor group (weight = KRW amount of each stock's net investment). Stock-selling skill is measured as a value-weighted average of ex-post buy-and-hold returns on stocks sold by an investor group (weight = KRW amount of each stock's net divestment). Notice that performance is measured over a quarter horizon and adjusted for stock characteristics (size and book-to-market ratio). The table is divided into two dimensions: size quintiles (in rows) and market liquidity (in columns). Size quintiles are updated each trading day and market liquidity is represented by number of shares purchased or sold by foreigners over total outstanding number of shares. Market is judged to be illiquid if the ratio is greater than 0.01 (1%). Standard errors are in the parentheses and numbers of observations are in the brackets.

Size Quintile	(no. of shares purchased or sold by foreigners) / (outstanding no. of shares)						Difference in Mean Tests		
	greater than 1%			lower than 1%			[7] = [6] – [4]	[8] = [6] – [5]	[9] = [3] - [6]
	[1] individual	[2] institution	[3] foreigner	[4] individual	[5] institution	[6] foreigner			
1 (small)	-0.035 (0.032) [152]	0.006 (0.043) [92]	-0.041 (0.032) [154]	0.021*** (0.007) [1024]	0.024*** (0.008) [1024]	-0.030*** (0.011) [991]	-0.051*** (0.013)	-0.054*** (0.013)	
2	0.030 (0.042) [292]	0.003 (0.024) [235]	0.031 (0.042) [292]	-0.011*** (0.004) [1024]	-0.010*** (0.004) [1024]	-0.021*** (0.010) [1021]	-0.010 (0.010)	-0.011 (0.010)	
3	-0.010 (0.019) [341]	-0.001 (0.019) [320]	-0.007 (0.019) [342]	-0.009*** (0.003) [1024]	-0.010*** (0.003) [1024]	0.000 (0.006) [1023]	0.010* (0.006)	0.011* (0.006)	
4	0.042** (0.020) [407]	0.030** (0.018) [398]	0.034** (0.019) [407]	-0.019*** (0.002) [1024]	-0.018*** (0.002) [1024]	0.009*** (0.004) [1024]	0.027*** (0.004)	0.027*** (0.004)	
5 (large)	-0.036** (0.016) [517]	-0.045*** (0.016) [517]	-0.045*** (0.016) [518]	-0.020*** (0.002) [1024]	-0.023*** (0.002) [1024]	0.003* (0.002) [1024]	0.023*** (0.003)	0.026*** (0.003)	-0.048*** (0.016)

Table 4
Performance Comparison
(Investment Horizon = one-half-year)

Table compares stock-picking skills among three investor-groups. Stock-picking skill is measured as a value-weighted average of stock-buying and stock-selling skills (weight = net investment and net divestment). Stock-buying skill, in return, is measured as a value-weighted average of ex-post buy-and-hold returns on stocks purchased by an investor group (weight = KRW amount of each stock's net investment). Stock-selling skill is measured as a value-weighted average of ex-post buy-and-hold returns on stocks sold by an investor group (weight = KRW amount of each stock's net divestment). Notice that performance is measured over a half-year horizon and adjusted for stock characteristics (size and book-to-market ratio). The table is divided into two dimensions: size quintiles (in rows) and market liquidity (in columns). Size quintiles are updated each trading day and market liquidity is represented by number of shares purchased or sold by foreigners over total outstanding number of shares. Market is judged to be illiquid if the ratio is greater than 0.01 (1%). Standard errors are in the parentheses and numbers of observations are in the brackets.

Size Quintile	(no. of shares purchased or sold by foreigners) / (outstanding no. of shares)						Difference in Mean Tests		
	greater than 1%			lower than 1%			[7] = [6] – [4]	[8] = [6] – [5]	[9] = [3] - [6]
	[1] individual	[2] institution	[3] foreigner	[4] individual	[5] institution	[6] foreigner			
1 (small)	-0.029 (0.054) [140]	0.042 (0.073) [84]	-0.035 (0.053) [142]	0.067*** (0.015) [952]	0.073*** (0.016) [952]	-0.045** (0.020) [914]	-0.111*** (0.025)	-0.117*** (0.026)	
2	0.004 (0.033) [276]	-0.015 (0.037) [225]	0.004 (0.033) [276]	0.029** (0.013) [952]	0.031** (0.014) [952]	0.013 (0.017) [949]	-0.016 (0.022)	-0.019 (0.022)	
3	0.068** (0.037) [329]	0.079** (0.038) [309]	0.069** (0.037) [330]	0.017** (0.008) [953]	0.020** (0.009) [953]	0.067*** (0.026) [951]	0.050** (0.027)	0.047** (0.028)	
4	0.028 (0.029) [391]	0.017 (0.028) [382]	0.016 (0.028) [391]	-0.029*** (0.004) [952]	-0.027*** (0.005) [952]	0.013** (0.007) [952]	0.042*** (0.008)	0.040*** (0.009)	
5 (large)	-0.049* (0.030) [477]	-0.073*** (0.029) [477]	-0.069*** (0.029) [478]	-0.042*** (0.003) [952]	-0.054*** (0.003) [952]	-0.005 (0.003) [952]	0.037*** (0.005)	0.049*** (0.005)	-0.064** (0.029)

Table 5
Performance Comparison
(Investment Horizon = one-year)

Table compares stock-picking skills among three investor-groups. Stock-picking skill is measured as a value-weighted average of stock-buying and stock-selling skills (weight = net investment and net divestment). Stock-buying skill, in return, is measured as a value-weighted average of ex-post buy-and-hold returns on stocks purchased by an investor group (weight = KRW amount of each stock's net investment). Stock-selling skill is measured as a value-weighted average of ex-post buy-and-hold returns on stocks sold by an investor group (weight = KRW amount of each stock's net divestment). Notice that performance is measured over a year horizon and adjusted for stock characteristics (size and book-to-market ratio). The table is divided into two dimensions: size quintiles (in rows) and market liquidity (in columns). Size quintiles are updated each trading day and market liquidity is represented by number of shares purchased or sold by foreigners over total outstanding number of shares. Market is judged to be illiquid if the ratio is greater than 0.01 (1%). Standard errors are in the parentheses and numbers of observations are in the brackets.

Size Quintile	(no. of shares purchased or sold by foreigners) / (outstanding no. of shares)						Difference in Mean Tests		
	greater than 1%			lower than 1%			[7] = [6] – [4]	[8] = [6] – [5]	[9] = [3] - [6]
	[1] individual	[2] institution	[3] foreigner	[4] individual	[5] institution	[6] foreigner			
1 (small)	0.119 (0.117) [117]	0.089 (0.134) [69]	0.111 (0.115) [119]	0.131*** (0.031) [808]	0.143*** (0.034) [808]	-0.021 (0.044) [765]	-0.152*** (0.054)	-0.165*** (0.055)	
2	-0.099*** (0.032) [246]	-0.106*** (0.028) [204]	-0.103*** (0.031) [246]	0.050*** (0.017) [808]	0.048*** (0.017) [808]	0.041 (0.029) [805]	-0.009 (0.034)	-0.007 (0.034)	
3	0.203*** (0.079) [269]	0.150** (0.067) [253]	0.202*** (0.081) [269]	0.003 (0.011) [808]	0.007 (0.013) [808]	0.108*** (0.037) [807]	0.105*** (0.039)	0.101*** (0.039)	
4	-0.082** (0.043) [320]	-0.072* (0.044) [312]	-0.093** (0.041) [320]	-0.064*** (0.007) [808]	-0.067*** (0.008) [808]	0.016 (0.014) [808]	0.080*** (0.015)	0.083*** (0.016)	
5 (large)	-0.212*** (0.056) [380]	-0.242*** (0.054) [380]	-0.234*** (0.054) [381]	-0.063*** (0.008) [808]	-0.102*** (0.007) [808]	-0.015** (0.009) [808]	0.048*** (0.012)	0.087*** (0.011)	-0.219*** (0.055)

Table 6
Performance Comparison
(Median)

As a robustness check, median performances, instead of mean performances, are compared among investor-groups.

Investment horizon	(no. of shares trade by foreigners) / (outstanding no. of shares) < 1%			
	Size quintile	individual	institution	foreigner
Month	1 (small)	-0.009	-0.009	-0.030
	5 (large)	-0.007	-0.007	0.000
Quarter	1 (small)	-0.017	-0.017	-0.060
	5 (large)	-0.017	-0.018	-0.003
Half-Year	1 (small)	-0.012	-0.010	-0.079
	5 (large)	-0.044	-0.050	-0.017
Year	1 (small)	-0.022	-0.024	-0.066
	5 (large)	-0.075	-0.084	-0.042

Table 7
Profitability of Tracking Foreigners

Stocks that were popular (or unpopular) among foreign investors are identified in three ways: (i) net purchase, (ii) net investment, and (iii) net purchase ratio. Net purchase is defined as number of shares purchased minus number of shares sold. Net investment is defined as net purchase multiplied by share price. Net purchase ratio is defined as net purchase divided by number of shares outstanding. Using each method, two portfolios are formed for each trading day: a portfolio comprising top-20 stocks and a portfolio comprising bottom-20 stocks. Value-weighted ex-post buy-and-hold return is then computed for the two portfolios. Strategies I, II, and III take a long position in top-20 stocks only. They do not take short positions. Each strategy respectively uses net purchase, net investment, and net purchase ratio to identify the top and the bottom portfolios. Strategies IV, V, and VI take a long position in top-20 stocks, while taking a short position in bottom-20 stocks. Return from each of these strategies is computed by combining the top-20 return and the bottom-20 return with a weight that reflects the relative size of the two portfolios.

PANEL A. Excess Return from Long Strategy				
	Month	Quarter	Half-Year	Year
Strategy I	0.0033** (0.0015) [1072]	-0.0016 (0.0029) [1024]	0.0032 (0.0043) [952]	-0.0090 (0.0095) [808]
Strategy II	0.0049*** (0.0013) [1072]	0.0080*** (0.0028) [1024]	0.0185*** (0.0041) [952]	-0.0033 (0.0087) [808]
Strategy III	0.0058*** (0.0020) [1072]	0.0085** (0.0041) [1024]	0.0028 (0.0061) [952]	-0.0286 (0.0139) [808]
PANEL B. Excess Return from Long combined with Short Strategy				
	Month	Quarter	Half-Year	Year
Strategy IV	0.0033*** (0.0012) [1071]	0.0031* (0.0023) [1023]	0.0119*** (0.0033) [951]	0.0090 (0.0078) [807]
Strategy V	0.0016* (0.0011) [1071]	0.0035* (0.0023) [1023]	0.0128*** (0.0032) [951]	0.0045 (0.0074) [807]
Strategy VI	0.0053*** (0.0016) [1071]	0.0160*** (0.0033) [1023]	0.0232*** (0.0049) [951]	0.0323*** (0.0119) [807]

Table 8
Profitability of Tracking Foreigners: Strategy VI by size

	Month	Quarter	Half-Year	Year
1 (small)	-0.0001 (0.0001) [310]	-0.0004** (0.0002) [277]	-0.0004* (0.0002) [255]	-0.0002* (0.0001) [216]
2	0.00001 (0.0001) [646]	-0.0001 (0.0001) [617]	0.0002 (0.0002) [590]	0.0004** (0.0002) [536]
3	-0.0002** (0.0001) [898]	-0.0003** (0.0001) [877]	0.0002 (0.0003) [832]	0.0005 (0.0005) [724]
4	0.0001 (0.0002) [1050]	0.0012*** (0.0003) [1005]	0.0028*** (0.0005) [936]	0.0031*** (0.0008) [798]
5 (large)	0.0055*** (0.0016) [1068]	0.0151*** (0.0033) [1020]	0.0202*** (0.0049) [948]	0.0285*** (0.0118) [804]